

A MOVING-BALL CLOSURE OF THE BAD-DENSITY STEP IN THE LEVEL-SET ROUTE TO SHARP SAINT-VENANT STABILITY

ABSTRACT. For a bounded open set $\Omega \subset \mathbb{R}^n$ of volume ω_n we prove the bounded sharp Saint-Venant stability estimate $\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega)$, where $\mathcal{A}$ is the Fraenkel asymmetry and δ_T the Saint-Venant deficit. The argument is built around the unscaled moving-ball distance $q(\rho) = \inf_z |E_\rho \Delta B_\rho(z)|$ on the torsion superlevel sets E_ρ . We first establish, on reduced boundaries and for almost every level, the exact level-set deficit identity expressing δ_T as a weighted integral of the Cauchy-Schwarz defect D_H and the isoperimetric defect D_I , and derive from it a profile-gap identity and a bad-density bound. Nestedness gives a perimeter-free Lipschitz bound for q ; we then prove a finite-perimeter one-sided affine first variation for q , a positive kinetic estimate, and an endpoint trace, which together close the stability estimate. The same-centre good-level package is obtained from the Fusco-Julin strong quantitative isoperimetric inequality and an elementary reduced-boundary radial trace lemma. The manuscript is self-contained modulo standard tools for sets of finite perimeter (the BV coarea formula, the Gauss-Green theorem, and Reshetnyak continuity) and the cited isoperimetric theorems.

CONTENTS

1. Introduction	1
2. Standing notation and scope	2
3. Finite-perimeter level-set identities	3
Regularity and level conventions	3
Coarea form of $-m'$ and the flux identity	4
The rearranged primitive	4
The exact deficit identity	5
4. Profile gap and bad-density estimate	6
5. Quantitative isoperimetry and same-centre trace control	7
6. Coarea differentiation and shell-admissible radii	11
Shell-admissible radii	12
7. Affine moving-ball first variation	13
8. Positive kinetic estimate	15
9. Endpoint trace and bounded stability	16
10. Proof architecture and constants	18
Appendix A. Standard tools from the theory of sets of finite perimeter	18
Appendix B. Dependency certificate	18
References	19

1. INTRODUCTION

Let $\Omega \subset \mathbb{R}^n$ ($n \geq 2$) be a bounded open set with $|\Omega| = \omega_n$ and let $u = u_\Omega$ be its torsion function. Writing $\delta_T(\Omega)$ for the Saint-Venant deficit and $\mathcal{A}(\Omega)$ for the Fraenkel asymmetry, the

main result is the bounded sharp stability estimate

$$\mathcal{A}(\Omega)^2 \leq C(n, R, \rho_*) \delta_T(\Omega) \quad (\Omega \subset B_R),$$

proved as Theorem 9.2. The strategy works directly with the torsion superlevel sets $E_\rho = \{u > t(\rho)\}$ of volume $\omega_n \rho^n$ and the *unscaled* moving-ball distance

$$q(\rho) := \inf_{z \in \mathbb{R}^n} |E_\rho \Delta B_\rho(z)|.$$

Because the family $\{E_\rho\}$ is nested, q is uniformly Lipschitz (Lemma 7.3); this is what lets the sparse set of bad-density radii be charged by Lebesgue measure rather than by the deficit.

The proof has three parts. Section 3 proves, on reduced boundaries and for a.e. level, the exact deficit identity $2\delta_T(\Omega) = \int_0^{\|u\|^\infty} (D_H + D_I) / (n^2 \omega_n^{2/n} m^{1-2/n}) dt$, where D_H is the Cauchy–Schwarz defect of $|\nabla u|$ on $\partial^* E_t$ and D_I the squared-perimeter isoperimetric defect, and Section 4 derives its profile-gap and bad-density consequences. Section 5 converts the Fusco–Julin strong quantitative isoperimetric inequality into a same-centre control of the homothetic trace. Sections 6–8 prove a finite-perimeter one-sided affine first variation for q and the resulting positive kinetic estimate, and Section 9 closes the argument through an endpoint trace. The one substantive external theorem is the Fusco–Julin quantitative isoperimetric inequality; the remaining ingredients are classical tools for sets of finite perimeter and classical rearrangement and isoperimetric inequalities, collected in Appendix A and the dependency certificate of Appendix B.

2. STANDING NOTATION AND SCOPE

Fix $n \geq 2$, $R \geq 1$, and $0 < \rho_* < 1$. Let $\Omega \subset B_R$ be bounded and open with $|\Omega| = \omega_n$, and let $u = u_\Omega \in H_0^1(\Omega)$ be the torsion function,

$$-\Delta u = 1 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega \quad (H_0^1 \text{ sense}).$$

We write the Saint–Venant deficit as

$$\delta_T(\Omega) := E(\Omega) - E(B_1) = -\frac{1}{2} \int_\Omega u + \frac{1}{2} \int_{B_1} u_{B_1} \geq 0.$$

Let $t(\rho)$ be the volume-radius parametrisation of the superlevels:

$$E_\rho := \{u > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n, \quad 0 < \rho < 1.$$

Set

$$w(\rho) := -t'(\rho), \quad d\mu(\rho) = w(\rho) d\rho.$$

For a.e. ρ , E_ρ has finite perimeter, $\mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0$, and the De Giorgi outer normal satisfies

$$\nu_\rho = -\frac{\nabla u}{|\nabla u|} \quad \mathcal{H}^{n-1}\text{-a.e. on } \partial^* E_\rho$$

(these level conventions are established in Lemma 3.1). On those levels define

$$V_\rho := \frac{w(\rho)}{|\nabla u|}, \quad H_{z,\rho}(x) := \frac{(x-z) \cdot \nu_\rho(x)}{\rho}, \quad \bar{V}_\rho := \frac{P(B_\rho)}{P(E_\rho)}.$$

The coarea relation of §3 (Lemma 3.2, with $w(\rho) \alpha(t(\rho)) = P(B_\rho)$) gives

$$(1) \quad \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = P(B_\rho), \quad \bar{V}_\rho = \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1}.$$

The two level-set defects are

$$D_H(t(\rho)) := \left(\int_{\partial^* E_\rho} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1} \right) \left(\int_{\partial^* E_\rho} |\nabla u| d\mathcal{H}^{n-1} \right) - P(E_\rho)^2,$$

$$D_I(t(\rho)) := P(E_\rho)^2 - P(B_\rho)^2.$$

Remark 2.1 (Scope). The bounded Saint–Venant estimate is deduced from the exact level-set identity (Theorem 3.11), proved in §3, and its profile-gap and bad-density consequences (Proposition 4.1). The same-centre good-level package (Theorem 5.4) rests on the theorem of Fusco–Julin, used in its exact same-centre form; the radial trace conversion from normal oscillation to the homothetic trace is Lemma 5.3. Every other step, including the first variation and the bad-density bookkeeping, is proved here.

3. FINITE-PERIMETER LEVEL-SET IDENTITIES

All integrals in this section are taken on De Giorgi reduced boundaries and all statements hold for almost every level; no smoothness of $\partial\Omega$, no graph parametrisation of the level sets, and no pointwise lower bound on $|\nabla u|$ are used. We keep the normalisation $|\Omega| = \omega_n$, so the ball of the same volume is B_1 ; the general case follows by scaling. Write

$$M := \|u\|_\infty, \quad L := |\Omega| = \omega_n, \quad m(t) := |\{u > t\}|,$$

$$\alpha(t) := \int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}, \quad \beta(t) := \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1}, \quad c_n := n^2 \omega_n^{2/n}.$$

By the Talenti pointwise comparison [10], $u^*(s) \leq u_{B_1}^*(s) = (1 - (s/\omega_n)^{2/n})/(2n)$, so $M \leq 1/(2n) < \infty$; here u^* is the decreasing rearrangement of u .

Regularity and level conventions.

Lemma 3.1 (Critical set and level conventions). *Let u be extended by 0 to $\mathbb{R}^n$. Then:*

- (a) $|\{u = 0\} \cap \Omega| = 0$ and $|\{\nabla u = 0\} \cap \Omega| = 0$;
- (b) for every $t \in (0, M)$, $|\{u = t\}| = 0$; hence m is continuous and strictly decreasing on $(0, M)$;
- (c) for a.e. $t \in (0, M)$: E_t has finite perimeter, $\partial^* E_t \subset \Omega$ up to $\mathcal{H}^{n-1}$ -null sets, $\nu_{E_t} = -\nabla u/|\nabla u|$ $\mathcal{H}^{n-1}$ -a.e. on $\partial^* E_t$, and $\mathcal{H}^{n-1}(\partial^* E_t \cap \{\nabla u = 0\}) = 0$.

Proof. (a) Testing $-\Delta u = 1$ against $u^- \in H_0^1(\Omega)$ gives $\int |\nabla u^-|^2 = -\int u^- \leq 0$, so $u \geq 0$ a.e.; the weak strong maximum principle on each connected component yields $u > 0$ a.e. in Ω [6], i.e. $|\{u = 0\} \cap \Omega| = 0$. Interior elliptic regularity gives $u \in W_{\text{loc}}^{2,p}(\Omega)$ for all $p < \infty$; applying Stampacchia's theorem [9] to the weak derivatives $\partial_i u$ gives $D^2 u = 0$ a.e. on $\{\nabla u = 0\}$, while $\Delta u = -1$ a.e. Hence $|\{\nabla u = 0\} \cap \Omega| = 0$.

(b) For $t \in (0, M)$ the Sobolev chain rule applied to $(u - t)^+$ gives $\nabla u = 0$ a.e. on $\{u = t\}$; since $\{u = t\} \subset \{u > 0\} \subset \Omega$ up to a null set, part (a) gives $|\{u = t\}| = 0$. A jump of m at t would have size $|\{u = t\}| = 0$, so m is continuous on $(0, M)$. For $0 < a < b < M$, some component of Ω has essential supremum of u exceeding b ; on it $\{a < u < b\}$ is a nonempty open set of positive measure, so $m(a) - m(b) = |\{a < u \leq b\}| > 0$.

(c) By the BV coarea formula [7, Thm. 13.1], [1, Thm. 3.40], for every Borel $\varphi \geq 0$,

$$(2) \quad \int_{\Omega} |\nabla u| \varphi dx = \int_{-\infty}^{\infty} \left(\int_{\partial^* E_t} \varphi d\mathcal{H}^{n-1} \right) dt.$$

Taking $\varphi = \mathbf{1}_{(0,M)}(u)$ (which equals 1 on $\partial^* E_t$ precisely for $t \in (0, M)$) gives $\int_0^M P(E_t) dt = \int_{\{0 < u < M\}} |\nabla u| dx \leq \int_{\Omega} |\nabla u| < \infty$, the last bound because $|\nabla u| \in L^2(\Omega) \subset L^1(\Omega)$ on the bounded set Ω ; hence $P(E_t) < \infty$ for a.e. t . For $t > 0$ the precise representative of u vanishes $\mathcal{H}^{n-1}$ -a.e. on $\mathbb{R}^n \setminus \Omega$ (as $u \in H_0^1(\Omega)$), and for a.e. t one has $\partial^* E_t \subset \{u = t\}$ up to $\mathcal{H}^{n-1}$ -null sets [1, 7]; therefore $\partial^* E_t \subset \Omega$ up to $\mathcal{H}^{n-1}$ -null sets. The orientation $\nu_{E_t} = -\nabla u / |\nabla u|$ is the standard one for superlevel sets of Sobolev functions [1]. Finally, (2) with $\varphi = \mathbf{1}_{\{\nabla u = 0\}}$ gives $\int_0^M \mathcal{H}^{n-1}(\partial^* E_t \cap \{\nabla u = 0\}) dt = \int_{\{\nabla u = 0\}} |\nabla u| dx = 0$, so the last assertion holds for a.e. t . $\square$

Statements “for a.e. t ” below refer to the full-measure set of levels furnished by Lemma 3.1(c).

Coarea form of $-m'$ and the flux identity.

Lemma 3.2 (Coarea form of $-m'$). *For a.e. $t \in (0, M)$, $-m'(t) = \alpha(t)$. In particular m is locally absolutely continuous on $(0, M)$ and $\alpha \in L_{\text{loc}}^1((0, M))$.*

Proof. Fix $0 < a < b < M$ and apply (2) to $\varphi = |\nabla u|^{-1} \mathbf{1}_{\{|\nabla u| > 0\}} \mathbf{1}_{(a,b)}(u)$. By Lemma 3.1(a) the integrand on the left equals $\mathbf{1}_{\{a < u < b\}}$ up to a null set, so $m(a) - m(b) = \int_a^b \alpha(t) dt$. Local absolute continuity of m and $-m' = \alpha$ a.e. follow. $\square$

Lemma 3.3 (Flux equals volume). *For a.e. $t \in (0, M)$, $\beta(t) = \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1} = m(t)$.*

Proof. For $\varepsilon > 0$ set $\phi_\varepsilon(s) := \min(1, \max(0, s/\varepsilon))$ and $\eta_\varepsilon := \phi_\varepsilon(u - t) \in H_0^1(\Omega)$. Testing $-\Delta u = 1$ against η_ε ,

$$\int_{\Omega} \nabla u \cdot \nabla \eta_\varepsilon dx = \int_{\Omega} \eta_\varepsilon dx.$$

The Sobolev chain rule [1, Thm. 3.99] gives $\nabla \eta_\varepsilon = \varepsilon^{-1} \nabla u \mathbf{1}_{\{t < u < t+\varepsilon\}}$, so the left side equals $\varepsilon^{-1} \int_{\{t < u < t+\varepsilon\}} |\nabla u|^2$, which by (2) (with $\varphi = |\nabla u| \mathbf{1}_{(t, t+\varepsilon)}(u)$) equals $\varepsilon^{-1} \int_t^{t+\varepsilon} \beta(\tau) d\tau$. As $\varepsilon \downarrow 0$, $\eta_\varepsilon \rightarrow \mathbf{1}_{E_t}$ boundedly (Lemma 3.1(b)), so the right side tends to $m(t)$. Since $\beta \in L^1((0, M))$ by (2) and the Dirichlet identity $\int_0^M \beta = \int_{\Omega} |\nabla u|^2 = \int_{\Omega} u < \infty$, at every Lebesgue point of β the left side tends to $\beta(t)$. Hence $\beta(t) = m(t)$ for a.e. t . $\square$

Remark 3.4 (Why a truncation, not a trace). Lemma 3.3 is the rigorous replacement, on an arbitrary finite-perimeter level set, for the formal divergence computation $m(t) = -\int_{E_t} \Delta u = -\int_{\partial^* E_t} \nabla u \cdot \nu_{E_t} = \beta(t)$ (using $\nabla u \cdot \nu_{E_t} = -|\nabla u|$); no boundary trace of ∇u on $\partial^* E_t$ is required [7, Ch. 17].

The rearranged primitive. Let $I(s) := \int_0^s u^*(\sigma) d\sigma$, $s \in [0, L]$.

Lemma 3.5 (Derivatives of the primitive). *I is absolutely continuous on $[0, L]$ with $I'(s) = u^*(s)$ a.e.; on every compact subinterval of $(0, L)$ the rearrangement u^* is absolutely continuous and*

$$(3) \quad I''(s) = (u^*)'(s) = \frac{1}{m'(u^*(s))} = -\frac{1}{\alpha(u^*(s))} \quad \text{for a.e. } s \in (0, L).$$

Proof. By the Talenti bound $u^* \in L^\infty(0, L) \subset L^1(0, L)$; the Lebesgue differentiation theorem gives $I' = u^*$ a.e. By Lemma 3.2, m is locally absolutely continuous with $m' = -\alpha$ a.e., and m is continuous and strictly decreasing on $(0, M)$ by Lemma 3.1(b), so $m(u^*(s)) = s$. The absolutely continuous inverse theorem for monotone functions [3, Thm. 0.4, §1] gives that u^* is absolutely continuous on compact subintervals of $(0, L)$ with $(u^*)'(s) = 1/m'(u^*(s))$ wherever $m'(u^*(s))$ exists and is nonzero. For a.e. $t \in (0, M)$ one has $0 < |E_t| < |\Omega|$, hence $P(E_t) > 0$ and $\alpha(t) > 0$; thus $\{m' = 0\} = \{\alpha = 0\}$ is null and carries no image measure ($|m(\{m' = 0\})| \leq \int_{\{m' = 0\}} |m'| = 0$). Since $m' = -\alpha$, (3) follows; absolute continuity of u^* on compact subintervals identifies the distributional and a.e. derivatives of $I' = u^*$. $\square$

Remark 3.6 (The corrected second derivative). The factor in (3) is $\alpha = \int_{\partial^* E_t} |\nabla u|^{-1}$, not $\beta = \int_{\partial^* E_t} |\nabla u|$. Since $\beta(t) = m(t)$ (Lemma 3.3), replacing α by β would give $I'' = -1/m$, which holds only when $|\nabla u|$ is constant on $\partial^* E_t$, i.e. on the ball; the discrepancy is exactly the Cauchy–Schwarz defect D_H introduced next, so the substitution would silently delete D_H from the identity.

The exact deficit identity. For a.e. $t \in (0, M)$ the two level-set defects of the standing notation read, as functions of the level,

$$D_H(t) = \alpha(t)\beta(t) - P(E_t)^2, \quad D_I(t) = P(E_t)^2 - c_n m(t)^{2-2/n},$$

consistent with §2 under $t = t(\rho)$ (so $P(B_\rho)^2 = c_n m(t)^{2-2/n}$). By Cauchy–Schwarz on $\partial^* E_t$ applied to $(|\nabla u|^{1/2}, |\nabla u|^{-1/2})$, $P(E_t)^2 \leq \alpha(t)\beta(t)$, so $D_H(t) \geq 0$, with equality iff $|\nabla u|$ is $\mathcal{H}^{n-1}$ -a.e. constant on $\partial^* E_t$. By the De Giorgi isoperimetric inequality [4, 7], $P(E_t) \geq n\omega_n^{1/n} m(t)^{1-1/n}$, so $D_I(t) \geq 0$, with equality iff E_t is equivalent to a ball.

Lemma 3.7 (Pointwise convexity formula). *Set $G(s) := I(s) + \frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}}$. Then for a.e. $s \in (0, L)$, with $t = u^*(s)$,*

$$G''(s) = \frac{1}{c_n s^{1-2/n}} - \frac{1}{\alpha(t)}.$$

Proof. Direct differentiation gives $\frac{d^2}{ds^2} \frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}} = \frac{1}{c_n s^{1-2/n}}$; add (3). $\square$

Lemma 3.8 (Convexity identity). $\int_0^L s G''(s) ds = \int_0^M \frac{D_H(t) + D_I(t)}{c_n m(t)^{1-2/n}} dt.$

Proof. Substitute $s = m(t)$, $ds = -\alpha(t) dt$, in $\int_0^L s G''(s) ds$ (Lemma 3.1(b) removes jump terms, and $\{\alpha = 0\}$ carries no image measure):

$$\int_0^L s G'' ds = \int_0^M m(t) \left(\frac{1}{c_n m(t)^{1-2/n}} - \frac{1}{\alpha(t)} \right) \alpha(t) dt = \int_0^M \left(\frac{m(t)\alpha(t)}{c_n m(t)^{1-2/n}} - m(t) \right) dt.$$

Use $m(t) = \beta(t)$ (Lemma 3.3) and add and subtract $P(E_t)^2/(c_n m(t)^{1-2/n})$:

$$\frac{m\alpha}{c_n m^{1-2/n}} - m = \frac{\alpha\beta - c_n m^{2-2/n}}{c_n m^{1-2/n}} = \frac{D_H + D_I}{c_n m^{1-2/n}}. \quad \square$$

Lemma 3.9 (Endpoint convexity-gap formula). *With $\Gamma(\Omega) := G'_-(L) - G(L)/L$, one has $\Gamma(\Omega) = \frac{1}{L} \int_0^L s G''(s) ds$.*

Proof. $G \in W_{\text{loc}}^{2,1}((0, L])$ and $G(0) = I(0) = 0$. Integration by parts on (ε, L) gives $\int_\varepsilon^L s G'' = L G'_-(L) - \varepsilon G'(\varepsilon) - G(L) + G(\varepsilon)$. By Lemma 3.5, $G'(s) = u^*(s) + s^{2/n}/(2n\omega_n^{2/n})$ is bounded, so $\varepsilon G'(\varepsilon) \rightarrow 0$ and $G(\varepsilon) \rightarrow 0$; divide by L . $\square$

Lemma 3.10 (Endpoint value). $\Gamma(\Omega) = \frac{2}{|\Omega|} (E(\Omega) - E(B_1)) = \frac{2}{\omega_n} \delta_T(\Omega).$

Proof. At $s = L$, $u^*(s) \rightarrow 0$ as $s \uparrow L$ (Lemma 3.1 gives $m(t) \uparrow L$ as $t \downarrow 0$), so $G'_-(L) = L^{2/n}/(2n\omega_n^{2/n})$ and $I(L) = \int_\Omega u dx$. With $L = \omega_n$, using $\frac{1}{2n} - \frac{1}{4+2n} = \frac{1}{2n} - \frac{1}{2(n+2)} = \frac{1}{n(n+2)}$,

$$\Gamma(\Omega) = \frac{L^{2/n}}{2n\omega_n^{2/n}} - \frac{L^{2/n}}{(4+2n)\omega_n^{2/n}} - \frac{1}{L} \int_\Omega u = \frac{1}{n(n+2)} - \frac{1}{\omega_n} \int_\Omega u.$$

For the unit-volume ball B_1 , $u_{B_1}(x) = (1 - |x|^2)/(2n)$ and

$$\int_{B_1} u_{B_1} = \frac{1}{2n} n \omega_n \int_0^1 (1 - r^2) r^{n-1} dr = \frac{\omega_n}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right) = \frac{\omega_n}{n(n+2)},$$

so $\frac{1}{\omega_n} \int_{B_1} u_{B_1} = \frac{1}{n(n+2)}$ and $\Gamma(B_1) = 0$ (the Pólya–Szegő identification [8, 2]). Subtracting, $\Gamma(\Omega) = \frac{1}{\omega_n} (\int_{B_1} u_{B_1} - \int_{\Omega} u) = \frac{2}{\omega_n} (E(\Omega) - E(B_1))$, since $E(\cdot) = -\frac{1}{2} \int(\cdot)$. $\square$

Theorem 3.11 (Exact deficit identity on reduced-boundary level sets). *Let $\Omega \subset \mathbb{R}^n$ be open with $|\Omega| = \omega_n$ and $u \in H_0^1(\Omega)$ the torsion function. Then (2) and Lemmas 3.2, 3.3, 3.5 hold, and*

$$(4) \quad \boxed{2 \delta_T(\Omega) = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt,}$$

with $D_H(t), D_I(t) \geq 0$ for a.e. t . By scaling, the corresponding identity $\frac{2}{|\Omega|} (E(\Omega) - E(B)) = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} (D_H + D_I) / (c_n m^{1-2/n}) dt$ holds for every open Ω of finite measure.

Proof. Combine Lemmas 3.8, 3.9, 3.10; the scaling statement follows since both sides are 0-homogeneous under $\Omega \mapsto \lambda\Omega$ after the displayed normalisation. $\square$

4. PROFILE GAP AND BAD-DENSITY ESTIMATE

Proposition 4.1 (Level-set budget and profile-gap bad set). *There are computable constants*

$$C_L = C_L(n, \rho_*), \quad C_\tau = C_\tau(n, \rho_*), \quad \kappa = \kappa(\rho_*) > 0,$$

such that, with

$$\rho_\delta := 1 - \kappa \sqrt{\delta_T(\Omega)}$$

and whenever $\rho_\delta \geq (1 + \rho_*)/2$, one has

$$(5) \quad \int_{\rho_*}^{\rho_\delta} (D_H(t(\rho)) + D_I(t(\rho))) d\mu(\rho) \leq C_L \delta_T(\Omega),$$

$$0 \leq w(\rho) \leq \frac{1}{n} \quad \text{for a.e. } \rho,$$

and, with

$$\tau_0 := \frac{\rho_*}{2n}, \quad B_\tau := \{\rho \in [\rho_*, \rho_\delta] : w(\rho) < \tau_0\}, \quad G_\tau := [\rho_*, \rho_\delta] \setminus B_\tau,$$

one has

$$(6) \quad |B_\tau| \leq C_\tau \delta_T(\Omega).$$

Proof. Recall $m(t) = |\{u > t\}|$ and $\alpha(t) = \int_{\partial^* \{u > t\}} |\nabla u|^{-1} d\mathcal{H}^{n-1}$ from §3. Theorem 3.11 gives the exact deficit identity

$$(7) \quad 2 \delta_T(\Omega) = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt,$$

all integrals taken on the reduced boundaries $\partial^* \{u > t\}$ and holding for a.e. level.

Changing variables $t = t(\rho)$, $m(t(\rho)) = \omega_n \rho^n$, and $dt = d\mu(\rho)$ in decreasing-radius orientation gives

$$2 \delta_T(\Omega) = \int_0^1 \frac{D_H(t(\rho)) + D_I(t(\rho))}{n^2 \omega_n \rho^{n-2}} d\mu(\rho).$$

Since $n \geq 2$ and $0 < \rho \leq 1$,

$$\int_{\rho_*}^{\rho_\delta} (D_H(t(\rho)) + D_I(t(\rho))) d\mu(\rho) \leq 2n^2\omega_n \delta_T(\Omega),$$

which proves (5), for instance with $C_L = 2n^2\omega_n$.

It remains to prove the density bounds. Write

$$a(\rho) := w(\rho) = -t'(\rho), \quad a_B(\rho) := \frac{\rho}{n}.$$

We next derive from (7) the profile derivative gap identity

$$(8) \quad 0 \leq a(\rho) \leq a_B(\rho) \quad \text{for a.e. } \rho \in (0, 1), \quad \int_0^1 \rho^n (a_B(\rho) - a(\rho)) d\rho = \frac{2}{\omega_n} \delta_T(\Omega).$$

For completeness, the algebra is as follows. Coarea gives

$$a(\rho) = \frac{n\omega_n \rho^{n-1}}{\alpha(t(\rho))}.$$

For the ball profile the corresponding coarea coefficient is

$$\alpha_B(\rho) = n^2\omega_n \rho^{n-2}, \quad \frac{n\omega_n \rho^{n-1}}{\alpha_B(\rho)} = \frac{\rho}{n} = a_B(\rho).$$

Since

$$D_H(t(\rho)) + D_I(t(\rho)) = \omega_n \rho^n (\alpha(t(\rho)) - \alpha_B(\rho)),$$

the nonnegativity of $D_H + D_I$ gives $0 \leq a \leq a_B$, and inserting the same formula in (7) gives the integral identity in (8).

Thus $0 \leq w(\rho) \leq \rho/n \leq 1/n$. On

$$B_\tau = \left\{ \rho \in [\rho_*, \rho_\delta] : a(\rho) < \frac{\rho_*}{2n} \right\}$$

one has

$$a_B(\rho) - a(\rho) \geq \frac{\rho}{n} - \frac{\rho_*}{2n} \geq \frac{\rho_*}{2n}.$$

Using $\rho^n \geq \rho_*^n$ on $[\rho_*, \rho_\delta]$ and (8),

$$\frac{\rho_*^{n+1}}{2n} |B_\tau| \leq \int_{B_\tau} \rho^n (a_B - a) d\rho \leq \frac{2}{\omega_n} \delta_T(\Omega).$$

Therefore (6) holds with

$$C_\tau = \frac{4n}{\omega_n \rho_*^{n+1}}.$$

Finally, choose for example $\kappa = (1 - \rho_*)/4$. Then $\rho_\delta \geq (1 + \rho_*)/2$ whenever $\delta_T(\Omega) \leq 1$, and in the large-deficit part of the final theorem this smallness is absorbed into the constant. $\square$

5. QUANTITATIVE ISOPERIMETRY AND SAME-CENTRE TRACE CONTROL

Theorem 5.1 (Fusco–Julin strong quantitative isoperimetry). *For every $n \geq 2$ there is a constant $C_{\text{FJ}}(n)$ with the following property. Let $E \subset \mathbb{R}^n$ be a set of finite perimeter and let $r_E > 0$ be defined by $|E| = \omega_n r_E^n$. Then there is a centre $z_E \in \mathbb{R}^n$ such that*

$$(9) \quad \left[\frac{|E \Delta B_{r_E}(z_E)|}{r_E^n} + \left(\frac{1}{r_E^{n-1}} \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z_E}{|x - z_E|} \right|^2 d\mathcal{H}^{n-1}(x) \right)^{1/2} \right]^2 \leq C_{\text{FJ}}(n) \frac{P(E) - P(B_{r_E})}{r_E^{n-1}}.$$

Proof. This is exactly [5, Theorem 1.1], with the Fusco–Julin asymmetry

$$A(E) = \min_z \left\{ \frac{|E \Delta B_{r_E}(z)|}{r_E^n} + \left(\frac{1}{r_E^{n-1}} \int_{\partial^* E} \left| \nu_E(x) - \nu_{B_{r_E}(z)}(\pi_{z,r_E}(x)) \right|^2 d\mathcal{H}^{n-1}(x) \right)^{1/2} \right\}$$

and their deficit $(P(E) - P(B_{r_E}))/r_E^{n-1}$. Since $\nu_{B_{r_E}(z)}(\pi_{z,r_E}(x)) = (x - z)/|x - z|$, their minimising centre is one centre for both the symmetric-difference term and the boundary-normal oscillation term. $\square$

Corollary 5.2 (Unnormalised same-centre consequences). *Fix $0 < \rho_* < 1$. Let $E \subset \mathbb{R}^n$ be a set of finite perimeter with $|E| = \omega_n \rho^n$, where $\rho \in [\rho_*, 1]$. Then there is a centre $z \in \mathbb{R}^n$ such that*

$$(10) \quad |E \Delta B_\rho(z)|^2 \leq C(n)(P(E) - P(B_\rho)),$$

and

$$(11) \quad \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1}(x) \leq C(n)(P(E) - P(B_\rho)).$$

Consequently, since $\rho \geq \rho_*$,

$$(12) \quad |E \Delta B_\rho(z)|^2 + \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1}(x) \leq C(n, \rho_*)(P(E)^2 - P(B_\rho)^2).$$

Proof. Apply Theorem 5.1 with $r_E = \rho$. Each summand in the left side of (9) is nonnegative, so

$$\frac{|E \Delta B_\rho(z)|^2}{\rho^{2n}} + \frac{1}{\rho^{n-1}} \int_{\partial^* E} \left| \nu_E - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1} \leq C(n) \frac{P(E) - P(B_\rho)}{\rho^{n-1}}.$$

Since $\rho \leq 1$, this gives (10) and (11). Finally,

$$P(E) - P(B_\rho) = \frac{P(E)^2 - P(B_\rho)^2}{P(E) + P(B_\rho)} \leq \frac{P(E)^2 - P(B_\rho)^2}{P(B_\rho)} \leq C(n, \rho_*)(P(E)^2 - P(B_\rho)^2).$$

$\square$

Lemma 5.3 (Oriented radial trace). *Fix $R \geq 1$ and $0 < \rho_* < 1$. Let $E \subset B_R$ be a set of finite perimeter with $|E| = \omega_n \rho^n$, where $\rho \in [\rho_*, 1]$, and let $z \in \mathbb{R}^n$ satisfy $|z| \leq R + 1$. Set*

$$e_z(x) := \frac{x - z}{|x - z|}, \quad W_z(x) := \left| \frac{|x - z|}{\rho} - 1 \right|.$$

Then

$$(13) \quad \int_{\partial^* E} W_z d\mathcal{H}^{n-1} \leq \frac{n}{\rho_*} |E \Delta B_\rho(z)| + C(R, \rho_*) \int_{\partial^* E} |\nu_E - e_z|^2 d\mathcal{H}^{n-1}.$$

Proof. Translate the notation temporarily so that $z = 0$, and write $B = B_\rho$, $r = |x|$, $e = x/r$. The point $x = 0$ is $\mathcal{H}^{n-1}$ -negligible on $\partial^* E$, so the value of e there is irrelevant. Define the bounded radial field

$$X(x) := \left| \frac{r}{\rho} - 1 \right| e.$$

For $0 < r < \rho$,

$$\operatorname{div} X = \frac{n-1}{r} - \frac{n}{\rho},$$

while for $r > \rho$,

$$\operatorname{div} X = \frac{n}{\rho} - \frac{n-1}{r}.$$

The field is singular only at the centre and has a Lipschitz radial coefficient away from the centre. To justify Gauss–Green, replace X by $\chi_\varepsilon(r)X$, where $\chi_\varepsilon = 0$ on B_ε and $\chi_\varepsilon = 1$ outside

$B_{2\varepsilon}$, smooth the radial profile, apply the classical finite-perimeter Gauss–Green formula [7], and let the smoothing and then $\varepsilon \downarrow 0$. The extra interior term $\int_E \nabla \chi_\varepsilon \cdot X \, dx$ is supported on the annulus $\{\varepsilon < |x| < 2\varepsilon\}$, where $|\nabla \chi_\varepsilon| \lesssim \varepsilon^{-1}$ and $|X|$ is bounded; since that annulus has volume $O(\varepsilon^n)$, this term is $O(\varepsilon^{n-1})$ and vanishes as $\varepsilon \downarrow 0$. The boundary error tends to zero because $\mathcal{H}^{n-1}(\partial^* E \cap B_{2\varepsilon}) \rightarrow \mathcal{H}^{n-1}(\partial^* E \cap \{0\}) = 0$, and the volume terms converge by dominated convergence, since r^{-1} is locally integrable in $\mathbb{R}^n$ for $n \geq 2$. Thus

$$\int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1} = \int_E \operatorname{div} X \, dx.$$

The same identity for B gives zero, because $X = 0$ on ∂B . Subtracting it yields

$$\int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1} = \int_{E \setminus B} \operatorname{div} X \, dx - \int_{B \setminus E} \operatorname{div} X \, dx.$$

On $E \setminus B$, $\operatorname{div} X \leq n/\rho$. On $B \setminus E$, $-\operatorname{div} X \leq n/\rho$. Hence

$$(14) \quad \int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1} \leq \frac{n}{\rho} |E \Delta B|.$$

On $\partial^* E$,

$$W_z = X \cdot \nu_E + W_z(1 - e_z \cdot \nu_E) = X \cdot \nu_E + \frac{W_z}{2} |\nu_E - e_z|^2.$$

Since $E \subset B_R$ and $|z| \leq R + 1$, every reduced-boundary point satisfies $|x - z| \leq 2R + 1$, and therefore

$$W_z(x) \leq 1 + \frac{2R + 1}{\rho_*}.$$

Combining this bound with (14) and using $\rho \geq \rho_*$ proves (13). $\square$

Theorem 5.4 (Same-centre good-level geometric package). *There exist constants*

$$\eta_I = \eta_I(n, R, \rho_*) > 0, \quad C_{\text{sc}} = C_{\text{sc}}(n, R, \rho_*),$$

such that the following holds. Define

$$G_I := \{\rho \in [\rho_*, \rho_\delta] : D_I(t(\rho)) \leq \eta_I\}.$$

For a.e. $\rho \in G_I$ there is a centre z_ρ^{sc} such that

$$(15) \quad |E_\rho \Delta B_\rho(z_\rho^{\text{sc}})|^2 + \left(\int_{\partial^* E_\rho} |H_{z_\rho^{\text{sc}}, \rho} - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \right)^2 \leq C_{\text{sc}} D_I(t(\rho)).$$

Proof. Let $\rho \in G_I$ be a finite-perimeter level. Apply Corollary 5.2 to E_ρ , and let $z = z_\rho^{\text{sc}}$ be the centre obtained there. Since

$$D_I(t(\rho)) = P(E_\rho)^2 - P(B_\rho)^2,$$

(12) gives

$$(16) \quad |E_\rho \Delta B_\rho(z)|^2 + \int_{\partial^* E_\rho} |\nu_\rho - e_z|^2 \, d\mathcal{H}^{n-1} \leq C(n, \rho_*) D_I(t(\rho)).$$

Choose η_I so small that the first term in (16) implies

$$|E_\rho \Delta B_\rho(z)| < |B_\rho|.$$

Then $E_\rho \cap B_\rho(z)$ has positive measure. Since $E_\rho \subset B_R$, this forces $B_\rho(z) \cap B_R \neq \emptyset$, and therefore

$$(17) \quad |z| \leq R + \rho \leq R + 1.$$

The radial trace lemma gives

$$\int_{\partial^* E_\rho} \left| \frac{|x-z|}{\rho} - 1 \right| d\mathcal{H}^{n-1} \leq C(n, R, \rho_*) (D_I(t(\rho))^{1/2} + D_I(t(\rho))).$$

Reducing $\eta_I \leq 1$, this is bounded by

$$(18) \quad C(n, R, \rho_*) D_I(t(\rho))^{1/2}.$$

For $H_{z,\rho} = (|x-z|/\rho)(e_z \cdot \nu_\rho)$,

$$|H_{z,\rho} - 1| \leq \left| \frac{|x-z|}{\rho} - 1 \right| + |1 - e_z \cdot \nu_\rho| = \left| \frac{|x-z|}{\rho} - 1 \right| + \frac{1}{2} |\nu_\rho - e_z|^2.$$

Together with (16) and (18), this gives

$$(19) \quad \int_{\partial^* E_\rho} |H_{z,\rho} - 1| d\mathcal{H}^{n-1} \leq C(n, R, \rho_*) D_I(t(\rho))^{1/2}.$$

Finally,

$$\bar{V}_\rho = \frac{P(B_\rho)}{P(E_\rho)}, \quad P(E_\rho) |1 - \bar{V}_\rho| = P(E_\rho) - P(B_\rho) \leq C(n, \rho_*) D_I(t(\rho)).$$

Combining this with (19) and again using $\eta_I \leq 1$,

$$\int_{\partial^* E_\rho} |H_{z,\rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \leq C(n, R, \rho_*) D_I(t(\rho))^{1/2}.$$

Squaring and adding the first estimate in (16) proves (15). $\square$

Lemma 5.5 (Velocity defect controlled by D_H). *For a.e. $\rho \in [\rho_*, \rho_\delta]$,*

$$(20) \quad \left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq D_H(t(\rho)).$$

Proof. Write $P = P(E_\rho)$, $P^* = P(B_\rho)$, $\alpha = \int_{\partial^* E_\rho} |\nabla u|^{-1} d\mathcal{H}^{n-1}$, and $\beta = \int_{\partial^* E_\rho} |\nabla u| d\mathcal{H}^{n-1}$. The coarea parametrisation gives $w\alpha = P^*$, and the flux identity gives $\beta = |E_\rho| = \omega_n \rho^n$. With $f := V_\rho = w/|\nabla u|$ and $\bar{f} := \bar{V}_\rho = P^*/P$,

$$\int_{\partial^* E_\rho} f d\mathcal{H}^{n-1} = P^*.$$

Moreover

$$\begin{aligned} \int_{\partial^* E_\rho} \frac{(f - \bar{f})^2}{f} d\mathcal{H}^{n-1} &= \int f d\mathcal{H}^{n-1} - 2\bar{f}P + \bar{f}^2 \int \frac{1}{f} d\mathcal{H}^{n-1} \\ &= -P^* + \frac{(P^*)^2}{P^2} \frac{\beta}{w} = -P^* + \frac{P^*}{P^2} \alpha\beta \\ &= \frac{P^*}{P^2} (\alpha\beta - P^2) = \frac{P^*}{P^2} D_H(t(\rho)). \end{aligned}$$

By Cauchy–Schwarz,

$$\left(\int |f - \bar{f}| d\mathcal{H}^{n-1} \right)^2 \leq \left(\int \frac{(f - \bar{f})^2}{f} d\mathcal{H}^{n-1} \right) \left(\int f d\mathcal{H}^{n-1} \right) = \left(\frac{P^*}{P} \right)^2 D_H(t(\rho)).$$

The isoperimetric inequality gives $P^* \leq P$, hence (20). $\square$

Lemma 5.6 (Superlevel asymmetry transfer). *Let $E \subset \Omega$, $|\Omega| = \omega_n$, and $|E| = \omega_n \rho^n$ with $\rho \in (0, 1]$. Then*

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E) + \frac{2}{\omega_n} |\Omega \setminus E|.$$

Here $\mathcal{A}(E)$ is the Fraenkel asymmetry of E relative to balls of volume $|E|$.

Proof. Let $s = |E|$, $L = |\Omega| = \omega_n$, and $\eta = L - s = |\Omega \setminus E|$. Choose z such that, for an arbitrary $\varepsilon > 0$,

$$|E \Delta B_\rho(z)| \leq s(\mathcal{A}(E) + \varepsilon).$$

Since $B_\rho(z) \subset B_1(z)$,

$$\begin{aligned} |\Omega \Delta B_1(z)| &\leq |\Omega \Delta E| + |E \Delta B_\rho(z)| + |B_\rho(z) \Delta B_1(z)| \\ &= \eta + |E \Delta B_\rho(z)| + \eta. \end{aligned}$$

Divide by L , use $s \leq L$, and let $\varepsilon \downarrow 0$. □

6. COAREA DIFFERENTIATION AND SHELL-ADMISSIBLE RADII

The first variation is proved on a fixed full-measure set of good-density radii. The point of the definition below is that all coarea differentiations are registered in advance by a countable family independent of the radius being tested.

Lemma 6.1 (Two-sided variable-thickness coarea differentiation). *Let $\xi \leq \zeta$ be bounded Borel functions on $\mathbb{R}^n$. Then for Lebesgue-a.e. positive level $s \in (0, \|u\|_\infty)$, with exceptional null set depending only on the pair (ξ, ζ) ,*

$$(21) \quad \lim_{h \downarrow 0} \frac{1}{h} |\{x : h\xi(x) < u(x) - s < h\zeta(x)\}| = \int_{\partial^* \{u > s\}} \frac{\zeta - \xi}{|\nabla u|} d\mathcal{H}^{n-1}.$$

The statement is used only at levels for which the right side is finite and $\mathcal{H}^{n-1}(\partial^ \{u > s\} \cap \{|\nabla u| = 0\}) = 0$.*

Proof. By Lemma 3.1(a), $|\{x \in \Omega : |\nabla u(x)| = 0\}| = 0$, so the critical set may be discarded in the volume computations below.

Suppose first that $\xi = \sum_j p_j \mathbf{1}_{A_j}$ and $\zeta = \sum_j q_j \mathbf{1}_{A_j}$ are simple over a common disjoint Borel partition, with $p_j \leq q_j$. By the BV coarea formula applied to $\mathbf{1}_{A_j} |\nabla u|^{-1} \mathbf{1}_{\{|\nabla u| > 0\}}$,

$$|\{hp_j < u - s < hq_j\} \cap A_j| = \int_{s+hp_j}^{s+hq_j} \int_{\partial^* \{u > \tau\}} \frac{\mathbf{1}_{A_j}}{|\nabla u|} d\mathcal{H}^{n-1} d\tau.$$

The interval of integration shrinks to s and has length $h(q_j - p_j)$; hence at a Lebesgue point of $\tau \mapsto \int_{\partial^* \{u > \tau\}} \mathbf{1}_{A_j} |\nabla u|^{-1} d\mathcal{H}^{n-1}$, division by h followed by $h \downarrow 0$ gives $(q_j - p_j) \int_{\partial^* \{u > s\}} \mathbf{1}_{A_j} |\nabla u|^{-1} d\mathcal{H}^{n-1}$. Summing over j proves (21) for simple pairs.

For bounded $\xi \leq \zeta$, fix $\eta > 0$ and choose simple pairs that sandwich the slab. Outer: simple $\sigma \leq \xi \leq \zeta \leq \sigma'$ with $\sigma' - \sigma \leq (\zeta - \xi) + 2\eta$; then $\{h\xi < u - s < h\zeta\} \subseteq \{h\sigma < u - s < h\sigma'\}$, so the lim sup of the left side of (21) is at most

$$\int_{\partial^* \{u > s\}} \frac{\sigma' - \sigma}{|\nabla u|} d\mathcal{H}^{n-1} \leq \int_{\partial^* \{u > s\}} \frac{\zeta - \xi}{|\nabla u|} d\mathcal{H}^{n-1} + 2\eta \int_{\partial^* \{u > s\}} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}.$$

Inner: simple $\sigma_* \geq \xi$, $\sigma'_* \leq \zeta$ with $\sigma_* \leq \sigma'_*$ and $\sigma'_* - \sigma_* \geq (\zeta - \xi - 2\eta)_+$; the reverse inclusion bounds the lim inf from below by

$$\int_{\partial^* \{u > s\}} \frac{(\zeta - \xi - 2\eta)_+}{|\nabla u|} d\mathcal{H}^{n-1}.$$

Letting $\eta \downarrow 0$ gives (21). The exceptional levels are fixed as follows: choose the outer and inner simple sandwiches for the dyadic sequence $\eta = 2^{-m}$, $m \geq 1$, by a fixed dyadic quantisation of the bounded functions ξ, ζ . The union of the exceptional null sets for those countably many simple pairs is still null and depends only on (ξ, ζ) . □

Shell-admissible radii. Let $\mathcal{U}$ be the countable family of finite unions of rational balls whose closures are compactly contained in Ω . This family is cofinal: for every compact $K \Subset \Omega$ there is $U \in \mathcal{U}$ with $K \subset U \Subset \Omega$. For

$$c, p, q, \lambda, r \in \mathbb{Q}, \quad -1 \leq c \leq 0, \quad p \leq q, \quad 1 \leq \lambda \leq \rho_*^{-1}, \quad r > 0, \quad z, a \in \mathbb{Q}^n,$$

and $U \in \mathcal{U}$, register the bounded Borel pairs

$$(p\mathbf{1}_U, q\mathbf{1}_U)$$

and

$$(\mathbf{1}_U(\min(c, \nabla u \cdot (\lambda(x - z) + a)) - r), \mathbf{1}_U(\max(c, \nabla u \cdot (\lambda(x - z) + a)) + r)).$$

This is a fixed countable family, independent of the radius being tested. We call $\rho \in G_\tau \cap (0, 1)$ *shell-admissible* if:

- (i) E_ρ has finite perimeter, $\partial^* E_\rho \subset \Omega$ up to $\mathcal{H}^{n-1}$ -null sets, and $\mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0$;
- (ii) t is differentiable at ρ , and

$$w(\rho) \int_{\partial^* E_\rho} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1} = P(B_\rho);$$

- (iii) $s = t(\rho)$ is a level where Lemma 6.1 holds for every registered pair above.

Almost every $\rho \in G_\tau \cap (0, 1)$ is shell-admissible. Indeed, each registered pair has an exceptional null set of levels s . If N is such a null set, then the one-dimensional area formula for the monotone absolutely continuous map t gives, on G_τ ,

$$|\{\rho : t(\rho) \in N\}| \leq \tau_0^{-1} \int_{\{t(\rho) \in N\}} w(\rho) d\rho \leq \tau_0^{-1} |N| = 0.$$

The remaining conditions hold for a.e. radius by BV coarea and the finite-perimeter level-set package. The critical-boundary condition follows from the preceding interior critical-set nullity and the coarea formula:

$$\int_0^{\|u\|_\infty} \mathcal{H}^{n-1}(\partial^* \{u > s\} \cap \{|\nabla u| = 0\}) ds = \int_{\{|\nabla u| = 0\}} |\nabla u| dx = 0.$$

Lemma 6.2 (Local BV deformation tail). *Let $E \subset B_R$ be a set of finite perimeter, let $\eta \in C_c^1(\mathbb{R}^n)$ satisfy $0 \leq \eta \leq 1$, and let $W \in C^1$ on a neighbourhood of a ball containing $E \cup \text{spt } \eta$. Set $T_h(x) = x + hW(x)$. Then*

$$\limsup_{h \rightarrow 0} \frac{1}{|h|} \int_{\mathbb{R}^n} \eta(x) |\mathbf{1}_{T_h(E)}(x) - \mathbf{1}_E(x)| dx \leq \int_{\partial^* E} \eta |W| d\mathcal{H}^{n-1}.$$

Proof. For $|h|$ small, T_h is a C^1 diffeomorphism on the fixed ball. Changing variables $x = T_h y$, it is enough, up to a multiplicative $1 + o(1)$ factor, to estimate

$$\int \eta(T_h y) |\mathbf{1}_E(T_h y) - \mathbf{1}_E(y)| dy.$$

Let $u_\varepsilon = \mathbf{1}_E * \varphi_\varepsilon$. For fixed h , the corresponding integrals with u_ε converge to the displayed integral because $u_\varepsilon \rightarrow \mathbf{1}_E$ in L^1 and composition with T_h has bounded Jacobian. For smooth u_ε ,

$$|u_\varepsilon(T_h y) - u_\varepsilon(y)| \leq |h| \int_0^1 |\nabla u_\varepsilon(y + \theta h W(y))| |W(y)| d\theta.$$

After integrating, change variables $x = y + \theta h W(y)$. These maps have Jacobians $1 + o(1)$ uniformly in θ . Thus, for fixed h ,

$$\frac{1}{|h|} \int \eta(T_h y) |u_\varepsilon(T_h y) - u_\varepsilon(y)| dy \leq (1 + o_h(1)) \int \psi_h d|Du_\varepsilon|,$$

where the continuous compactly supported weights ψ_h are uniformly bounded and satisfy $\psi_h \rightarrow \eta|W|$ uniformly as $h \rightarrow 0$. Letting $\varepsilon \downarrow 0$ at fixed h , strict BV_{loc} convergence of the standard mollifications and Reshetnyak continuity for continuous weights [1, 7] give the same bound with $d|D\mathbf{1}_E|$. Taking $\limsup_{h \rightarrow 0}$ then gives

$$\limsup_{h \rightarrow 0} \frac{1}{|h|} \int \eta(x) |\mathbf{1}_{T_h(E)}(x) - \mathbf{1}_E(x)| dx \leq \int \eta|W| d|D\mathbf{1}_E|.$$

Since $|D\mathbf{1}_E| = \mathcal{H}^{n-1} \llcorner \partial^* E$, this is the claim. $\square$

7. AFFINE MOVING-BALL FIRST VARIATION

Lemma 7.1 (Affine shell estimate). *Let ρ be shell-admissible. Fix $z, a \in \mathbb{R}^n$, and define*

$$T_h(x) := z + ha + \left(1 + \frac{h}{\rho}\right)(x - z) = x + h \left(\frac{x - z}{\rho} + a\right).$$

Then

$$(22) \quad \limsup_{h \downarrow 0} \frac{|E_{\rho+h} \Delta T_h(E_\rho)|}{h} \leq \int_{\partial^* E_\rho} |V_\rho - H_{z,\rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

Proof. Let $E = E_\rho$, $W(x) = (x - z)/\rho + a$, and $w = w(\rho)$. Since $\Omega \subset B_R$, W is bounded on Ω . The finite Radon measure

$$\left(\frac{w}{|\nabla u|} + |W| + \left| \frac{w}{|\nabla u|} - W \cdot \nu_\rho \right| \right) \mathcal{H}^{n-1} \llcorner \partial^* E$$

is finite. Fix $\varepsilon > 0$. By inner regularity and $\partial^* E \subset \Omega$ up to $\mathcal{H}^{n-1}$ -null sets, choose a compact $K \subset \partial^* E \cap \Omega$ such that this measure of $\partial^* E \setminus K$ is at most ε . Then choose $U \in \mathcal{U}$ with $K \subset U \Subset \Omega$.

We first estimate the symmetric difference in U . On a neighbourhood of $\bar{U}$, interior elliptic regularity gives $u \in C^1$, and uniformly for $x \in U$,

$$T_h^{-1}(x) = x - hW(x) + O(h^2), \quad u(T_h^{-1}x) = u(x) - h\nabla u(x) \cdot W(x) + o(h).$$

Also $t(\rho + h) = t(\rho) - hw + o(h)$. Set

$$F(x) := w + \nabla u(x) \cdot W(x).$$

For $x \in U$,

$$x \in E_{\rho+h} \iff u(x) > t(\rho) - hw + o(h),$$

while

$$x \in T_h(E) \iff u(x) > t(\rho) + h\nabla u(x) \cdot W(x) + o(h).$$

Thus the part of the symmetric difference in U is contained, up to a uniform $o(h)$ outward error, in the two-sided slab whose endpoints are $\min(-w, \nabla u \cdot W)$ and $\max(-w, \nabla u \cdot W)$. Precisely, fix $\delta > 0$. Choose rational c, λ, z_0, a_0 with $c \in [-1, 0]$, $1 \leq \lambda \leq \rho_*^{-1}$, and

$$|c + w| + \sup_{\bar{U}} |\nabla u \cdot (\lambda(x - z_0) + a_0) - \nabla u \cdot W| \leq \delta.$$

Then choose a rational buffer $r > 2\delta$. For all sufficiently small $h > 0$, the local symmetric difference is contained in the registered slab with endpoints

$$\min(c, \nabla u \cdot (\lambda(x - z_0) + a_0)) - r, \quad \max(c, \nabla u \cdot (\lambda(x - z_0) + a_0)) + r.$$

Lemma 6.1, followed by $\delta, r \downarrow 0$, gives

$$\limsup_{h \downarrow 0} \frac{|(E_{\rho+h} \Delta T_h(E)) \cap U|}{h} \leq \int_{\partial^* E \cap U} \frac{|F|}{|\nabla u|} d\mathcal{H}^{n-1}.$$

On the complement use

$$E_{\rho+h} \Delta T_h(E) \subset (E_{\rho+h} \Delta E) \cup (E \Delta T_h(E)).$$

For the level-set part, the exact volume derivative and the registered localized one-sided coarea formula on U give the complement estimate. Indeed,

$$|E_{\rho+h} \setminus E| = \omega_n((\rho + h)^n - \rho^n) = h P(B_\rho) + o(h) = h w \int_{\partial^* E} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1} + o(h),$$

where the last equality is the shell-admissible flux identity. On U , the same rational-buffer use of Lemma 6.1, now with one endpoint 0, gives

$$\lim_{h \downarrow 0} \frac{|(E_{\rho+h} \setminus E) \cap U|}{h} = w \int_{\partial^* E \cap U} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}.$$

Subtracting the local shell from the exact total shell yields

$$\limsup_{h \downarrow 0} \frac{|(E_{\rho+h} \Delta E) \setminus U|}{h} \leq w \int_{\partial^* E \setminus U} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1} \leq w \int_{\partial^* E \setminus K} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}.$$

For the affine deformation part, choose $\eta \in C_c^1(\mathbb{R}^n)$ with $\eta = 0$ on a neighbourhood $N_K \Subset U$ of K , $\eta = 1$ on a fixed neighbourhood of $\overline{B_{R+1}} \setminus U$, and $0 \leq \eta \leq 1$. Since $E \subset B_R$ and W is bounded, $T_h(E) \subset B_{R+1}$ for all small $h > 0$; hence $(E \Delta T_h(E)) \setminus U \subset \overline{B_{R+1}} \setminus U \subset \{\eta = 1\}$. As $\eta = 0$ near K and $0 \leq \eta \leq 1$, Lemma 6.2 gives

$$\limsup_{h \downarrow 0} \frac{|(E \Delta T_h(E)) \setminus U|}{h} \leq \int_{\partial^* E} \eta |W| d\mathcal{H}^{n-1} \leq \int_{\partial^* E \setminus K} |W| d\mathcal{H}^{n-1}.$$

The two tail terms are bounded by 2ε . Therefore

$$\limsup_{h \downarrow 0} \frac{|E_{\rho+h} \Delta T_h(E_\rho)|}{h} \leq \int_{\partial^* E_\rho \cap U} \frac{|F|}{|\nabla u|} d\mathcal{H}^{n-1} + 2\varepsilon \leq \int_{\partial^* E_\rho} \frac{|F|}{|\nabla u|} d\mathcal{H}^{n-1} + 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$, and using

$$\frac{F}{|\nabla u|} = \frac{w}{|\nabla u|} + \frac{\nabla u \cdot W}{|\nabla u|} = V_\rho - W \cdot \nu_\rho = V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho$$

on $\partial^* E_\rho$, proves the estimate. $\square$

Theorem 7.2 (One-sided first variation of q). *For every $\rho \in (0, 1]$, define*

$$q(\rho) := \inf_{z \in \mathbb{R}^n} |E_\rho \Delta B_\rho(z)|.$$

At every shell-admissible $\rho \in (0, 1)$,

$$(23) \quad D^+ q(\rho) := \limsup_{h \downarrow 0} \frac{q(\rho + h) - q(\rho)}{h} \leq \frac{n}{\rho} q(\rho) + \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1},$$

where z_ρ is any minimiser of $z \mapsto |E_\rho \Delta B_\rho(z)|$.

Proof. The minimiser exists. Indeed, $z \mapsto |E_\rho \Delta B_\rho(z)|$ is continuous by translation continuity in L^1 , and if $|z| > R + \rho$, then $B_\rho(z) \cap E_\rho = \emptyset$, so the value is $2|E_\rho|$. Thus the infimum is attained in a closed ball.

Fix a minimiser z_ρ and $a \in \mathbb{R}^n$. For the affine map in Lemma 7.1,

$$T_h(B_\rho(z_\rho)) = B_{\rho+h}(z_\rho + ha).$$

Therefore

$$\begin{aligned} q(\rho + h) &\leq |E_{\rho+h} \Delta T_h(B_\rho(z_\rho))| \\ &\leq |E_{\rho+h} \Delta T_h(E_\rho)| + |T_h(E_\rho) \Delta T_h(B_\rho(z_\rho))|. \end{aligned}$$

Since T_h is injective affine with constant Jacobian $(1 + h/\rho)^n$,

$$|T_h(E_\rho) \Delta T_h(B_\rho(z_\rho))| = \left(1 + \frac{h}{\rho}\right)^n q(\rho).$$

Subtract $q(\rho)$, divide by h , take the limsup as $h \downarrow 0$, and use Lemma 7.1. Then take the infimum over a . $\square$

Lemma 7.3 (Unconditional Lipschitz bound for q). *For $0 < \rho \leq \sigma \leq 1$,*

$$|q(\sigma) - q(\rho)| \leq 2\omega_n(\sigma^n - \rho^n) \leq 2n\omega_n|\sigma - \rho|.$$

Consequently q is absolutely continuous and differentiable a.e.

Proof. Use the same centre for the two radii and the nestedness $E_\rho \subset E_\sigma$, $B_\rho(z) \subset B_\sigma(z)$:

$$q(\sigma) \leq q(\rho) + |E_\sigma \setminus E_\rho| + |B_\sigma \setminus B_\rho| = q(\rho) + 2\omega_n(\sigma^n - \rho^n).$$

Interchange ρ and σ to get the reverse inequality. $\square$

8. POSITIVE KINETIC ESTIMATE

Lemma 8.1 (Transfer to a distance-minimising centre). *After reducing $\eta_I = \eta_I(n, R, \rho_*)$ if necessary, the following holds. Let $\rho \in G_I$ be a level where Theorem 5.4 holds, and let $z_\rho^{\min}$ minimise $z \mapsto |E_\rho \Delta B_\rho(z)|$. Then*

$$(24) \quad \left(\int_{\partial^* E_\rho} |H_{z_\rho^{\min}, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*) D_I(t(\rho)).$$

Proof. Let z_ρ^{sc} be the centre from Theorem 5.4. Minimality gives

$$|E_\rho \Delta B_\rho(z_\rho^{\min})| \leq |E_\rho \Delta B_\rho(z_\rho^{\text{sc}})| \leq C D_I(t(\rho))^{1/2}.$$

Thus

$$|B_\rho(z_\rho^{\min}) \Delta B_\rho(z_\rho^{\text{sc}})| \leq C D_I(t(\rho))^{1/2}.$$

For equal balls with centre distance d ,

$$|B_\rho(z_1) \Delta B_\rho(z_2)| \geq c(n) \rho^{n-1} \min\{d, \rho\}.$$

Indeed, in coordinates $x = z_1 + s e + y$ with $e = (z_2 - z_1)/d$ and $y \perp e$, one has $B_\rho(z_1) \setminus B_\rho(z_2) = \{s^2 + |y|^2 < \rho^2, s \leq d/2\}$; integrating the $(n-1)$ -dimensional cross-sections over $s \in [0, \frac{1}{2} \min\{d, \rho\}]$, where $\rho^2 - s^2 \geq \frac{3}{4}\rho^2$, gives $|B_\rho(z_1) \setminus B_\rho(z_2)| \geq \frac{1}{2}\omega_{n-1}(3/4)^{(n-1)/2} \rho^{n-1} \min\{d, \rho\}$, and the symmetric difference is twice this. Choose η_I so small that the preceding upper bound forces the linear branch $d < \rho$. Since $\rho \geq \rho_*$,

$$|z_\rho^{\min} - z_\rho^{\text{sc}}| \leq C(n, \rho_*) D_I(t(\rho))^{1/2}.$$

The homothetic trace is affine in the centre:

$$H_{z_\rho^{\min}, \rho} - H_{z_\rho^{\text{sc}}, \rho} = -\frac{(z_\rho^{\min} - z_\rho^{\text{sc}}) \cdot \nu_\rho}{\rho}.$$

On G_I , the perimeter is uniformly bounded because $D_I(t(\rho)) \leq \eta_I$ and $\rho \in [\rho_*, 1]$. Hence

$$\int_{\partial^* E_\rho} |H_{z_\rho^{\min}, \rho} - H_{z_\rho^{\text{sc}}, \rho}| d\mathcal{H}^{n-1} \leq C(n, R, \rho_*) D_I(t(\rho))^{1/2}.$$

Combining this with (15) via the triangle inequality,

$$\int_{\partial^* E_\rho} |H_{z_\rho^{\min}, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \leq \int_{\partial^* E_\rho} |H_{z_\rho^{\min}, \rho} - H_{z_\rho^{\text{sc}}, \rho}| d\mathcal{H}^{n-1} + \int_{\partial^* E_\rho} |H_{z_\rho^{\text{sc}}, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \leq C(n, R, \rho_*) D_I(t(\rho))^{1/2},$$

and squaring gives (24). $\square$

Proposition 8.2 (Positive kinetic bound). *Let $J \subset [\rho_*, \rho_\delta]$ be an interval. Then*

$$\int_J (q'_+(\rho))^2 d\rho \leq C(n, R, \rho_*) \delta_T(\Omega),$$

where $q'_+(\rho) = \max\{q'(\rho), 0\}$ at a.e. differentiability points.

Proof. Let

$$G := J \cap G_I \cap G_\tau, \quad \mathcal{B} := J \setminus G.$$

Since q is Lipschitz, q' exists a.e.; shell-admissible radii have full measure in G_τ . At points where both properties hold, $D^+q = q'$. The right side of Theorem 7.2 is nonnegative, so the theorem also bounds q'_+ . On G , outside this null set, Theorem 7.2, Theorem 5.4, Lemma 8.1, and Lemma 5.5 give

$$(q'_+)^2 \leq C(n, R, \rho_*) (D_H(t(\rho)) + D_I(t(\rho))).$$

Indeed $q^2 \leq CD_I$ on G_I , and

$$\inf_a \int |V_\rho - H_{z_\rho^{\min}, \rho} - a \cdot \nu_\rho| \leq \int |V_\rho - \bar{V}_\rho| + \int |H_{z_\rho^{\min}, \rho} - \bar{V}_\rho|.$$

On G_τ , $d\rho \leq \tau_0^{-1} d\mu$. Hence the budget (5) yields

$$\int_G (q'_+)^2 d\rho \leq C \delta_T(\Omega).$$

It remains to pay $\mathcal{B}$. By Lemma 7.3, $q'_+ \leq 2n\omega_n$ a.e. The bad-density part has $|J \cap B_\tau| \leq C \delta_T(\Omega)$. For the bad-isoperimetric part on good density,

$$|J \cap G_\tau \cap G_I^c| \leq \tau_0^{-1} \mu(G_I^c) \leq \frac{C}{\eta_I} \int_J D_I(t(\rho)) d\mu(\rho) \leq C \delta_T(\Omega).$$

Therefore $|\mathcal{B}| \leq C \delta_T(\Omega)$, and

$$\int_{\mathcal{B}} (q'_+)^2 d\rho \leq C \delta_T(\Omega). \quad \square$$

9. ENDPOINT TRACE AND BOUNDED STABILITY

Lemma 9.1 (Moving-ball endpoint trace). *Let*

$$L_0 := \frac{1 - \rho_*}{8}, \quad J := [\rho_\delta - L_0, \rho_\delta].$$

Assume $\delta_T(\Omega)$ is small enough that $\rho_\delta \geq (1 + \rho_*)/2$. Then $J \subset [\rho_*, \rho_\delta]$ and

$$q(\rho_\delta)^2 \leq C(n, R, \rho_*) \delta_T(\Omega).$$

Proof. The inclusion $J \subset [\rho_*, \rho_\delta]$ follows from $\rho_\delta - \rho_* \geq (1 - \rho_*)/2$. On $G = J \cap G_I \cap G_\tau$,

$$q(\rho)^2 \leq CD_I(t(\rho)), \quad d\rho \leq \tau_0^{-1} d\mu(\rho),$$

so

$$\int_G q(\rho)^2 d\rho \leq C \delta_T(\Omega).$$

On $J \setminus G$, the measure is $O(\delta_T)$ by the proof of Proposition 8.2, while $q \leq 2\omega_n$. Hence

$$\int_J q(\rho)^2 d\rho \leq C \delta_T(\Omega).$$

There is therefore $\rho_0 \in J$ with

$$q(\rho_0)^2 \leq CL_0^{-1} \delta_T(\Omega).$$

Since q is absolutely continuous and $\rho_0 \leq \rho_\delta$,

$$q(\rho_\delta) \leq q(\rho_0) + \int_{\rho_0}^{\rho_\delta} q'_+(\rho) d\rho.$$

By Cauchy–Schwarz and Proposition 8.2,

$$q(\rho_\delta) \leq C\delta_T(\Omega)^{1/2} + L_0^{1/2} \left(\int_J (q'_+)^2 d\rho \right)^{1/2} \leq C\delta_T(\Omega)^{1/2}.$$

□

Theorem 9.2 (Bounded sharp Saint–Venant stability). *There are computable constants $C = C(n, R, \rho_*)$ and $\delta_0 = \delta_0(n, R, \rho_*) > 0$, depending only on the level-set identity constants, the Fusco–Julin constant, and the explicit constants above, such that every bounded open $\Omega \subset B_R$, $|\Omega| = \omega_n$, satisfies*

$$\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega).$$

Proof. Let κ be the constant fixed in Proposition 4.1. First assume $\delta_T(\Omega) \leq \delta_0$, with δ_0 chosen so that $\rho_\delta = 1 - \kappa\sqrt{\delta_T(\Omega)} \geq (1 + \rho_*)/2$, $\delta_T(\Omega) \leq 1$, and the smallness thresholds used in Theorem 5.4 and Lemma 8.1 hold. Lemma 9.1 gives

$$q(\rho_\delta)^2 \leq C\delta_T(\Omega).$$

Since $|E_{\rho_\delta}| = \omega_n \rho_\delta^n$ and $\rho_\delta \geq \rho_*$,

$$\mathcal{A}(E_{\rho_\delta})^2 \leq \frac{q(\rho_\delta)^2}{\omega_n^2 \rho_\delta^{2n}} \leq C(n, \rho_*) \delta_T(\Omega).$$

Moreover $E_{\rho_\delta} \subset \Omega$, and by the definition of ρ_δ ,

$$|\Omega \setminus E_{\rho_\delta}| = \omega_n(1 - \rho_\delta^n) \leq n\omega_n(1 - \rho_\delta) = n\omega_n \kappa \delta_T(\Omega)^{1/2}.$$

Lemma 5.6 gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\rho_\delta}) + C(n, \rho_*) \delta_T(\Omega)^{1/2}.$$

Squaring proves the estimate in the small-deficit regime.

If $\delta_T(\Omega) > \delta_0$, then $\mathcal{A}(\Omega) < 2$, so

$$\mathcal{A}(\Omega)^2 \leq \frac{4}{\delta_0} \delta_T(\Omega).$$

Enlarging C completes the proof.

□

10. PROOF ARCHITECTURE AND CONSTANTS

The argument is organised around the following five steps.

- (1) The exact level-set deficit identity (Theorem 3.11) writes $\delta_T(\Omega)$ as a weighted reduced-boundary integral of the Cauchy–Schwarz defect D_H and the isoperimetric defect D_I ; the profile-gap identity and the bad-density bound of Proposition 4.1 are its ρ -parametrised consequences.
- (2) The affine shell estimate (Lemma 7.1) is a finite-perimeter limsup estimate, proved by localized coarea differentiation on variable slabs (Lemma 6.1) together with the local BV deformation-tail lemma (Lemma 6.2). Its only bounded-variation input is Reshetnyak continuity for a continuous weight, applied as an *upper* bound to a swept-volume tail made arbitrarily small by inner regularity; no lower-semicontinuity passage is used.
- (3) The endpoint trace (Lemma 9.1) uses the upper Dini derivative D^+q together with the global Lipschitz bound for q (Lemma 7.3).
- (4) Bad-density levels are charged by Lebesgue measure rather than by the deficit, which is possible because q is uniformly Lipschitz by nestedness.
- (5) The same-centre good-level package (Theorem 5.4) is obtained from the Fusco–Julin strong quantitative isoperimetric inequality (Theorem 5.1); the passage from normal oscillation to the homothetic trace is the radial trace Lemma 5.3.

All constants in Theorem 9.2 depend only on n , R , and ρ_* , together with the Fusco–Julin constant $C_{\text{FJ}}(n)$. Tracing them through the proof: $C_L = 2n^2\omega_n$ and $C_\tau = 4n/(\omega_n\rho_*^{n+1})$ (Proposition 4.1); $\kappa = (1-\rho_*)/4$ and $\tau_0 = \rho_*/(2n)$; the same-centre constant $C_{\text{sc}}(n, R, \rho_*)$ and threshold $\eta_I(n, R, \rho_*)$ (Theorem 5.4, Lemma 8.1); the ball–ball lower constant $\frac{1}{2}\omega_{n-1}(3/4)^{(n-1)/2}$; the slab length $L_0 = (1-\rho_*)/8$; and the large-deficit cutoff $4/\delta_0$. No constant depends on Ω beyond these parameters.

APPENDIX A. STANDARD TOOLS FROM THE THEORY OF SETS OF FINITE PERIMETER

For the reader’s convenience we record the three classical results used above, in the exact form needed; all hold for sets of finite perimeter with no graph or manifold regularity.

- (1) *BV coarea formula* [7, Thm. 13.1], [1, Thm. 3.40]: for $u \in BV_{\text{loc}}(\mathbb{R}^n)$ and Borel $\varphi \geq 0$, $\int |\nabla u| \varphi = \int_{\mathbb{R}} (\int_{\partial^* \{u>t\}} \varphi d\mathcal{H}^{n-1}) dt$. This underlies (2) and Lemma 6.1.
- (2) *Gauss–Green on sets of finite perimeter* [7, Ch. 17]: for a bounded Lipschitz vector field X and a set E of finite perimeter, $\int_E \text{div } X = \int_{\partial^* E} X \cdot \nu_E d\mathcal{H}^{n-1}$. This is the form used in Lemma 5.3.
- (3) *Strict BV convergence and Reshetnyak continuity* [1, Thm. 2.39], [7, Ch. 20]: if $v_k \rightarrow v$ in L^1_{loc} with $|Dv_k|(\mathbb{R}^n) \rightarrow |Dv|(\mathbb{R}^n)$, then $\int \psi d|Dv_k| \rightarrow \int \psi d|Dv|$ for every continuous compactly supported ψ . Applied to mollifications $v_k = \mathbf{1}_E * \varphi_{1/k}$, this gives the upper bound in Lemma 6.2.

APPENDIX B. DEPENDENCY CERTIFICATE

Every step is either proved in this manuscript or quoted from a named published theorem.

Proved internally: the exact deficit identity and its inputs (coarea form of $-m'$, the Lipschitz-truncation flux identity, the rearranged primitive with $I'' = -1/\alpha$) in §3; the profile-gap and bad-density estimates in §4; the oriented radial trace, the same-centre package, the velocity defect, and the superlevel asymmetry transfer in §5; the two-sided coarea differentiation lemma,

the shell-admissible registration, and the local BV deformation tail in §6; the affine shell estimate, the one-sided first variation, and the Lipschitz bound for q in §7; the centre transfer and the positive kinetic estimate in §8; the endpoint trace and the bounded stability theorem in §9.

Quoted with exact citation: the Fusco–Julin strong quantitative isoperimetric inequality [5]; the BV coarea formula, the finite-perimeter Gauss–Green theorem, and Reshetnyak continuity [1, 7]; the De Giorgi isoperimetric inequality [4]; the Talenti rearrangement L^∞ bound [10]; the absolutely continuous monotone inverse theorem [3]; Stampacchia’s theorem on $\{\nabla u = 0\}$ [9]; interior elliptic regularity and the weak maximum principle [6]; and the Pólya–Szegő identification of the ball energy [8, 2]. Every ingredient is therefore either proved above or quoted from the published literature; the argument can be checked from this manuscript alone.

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